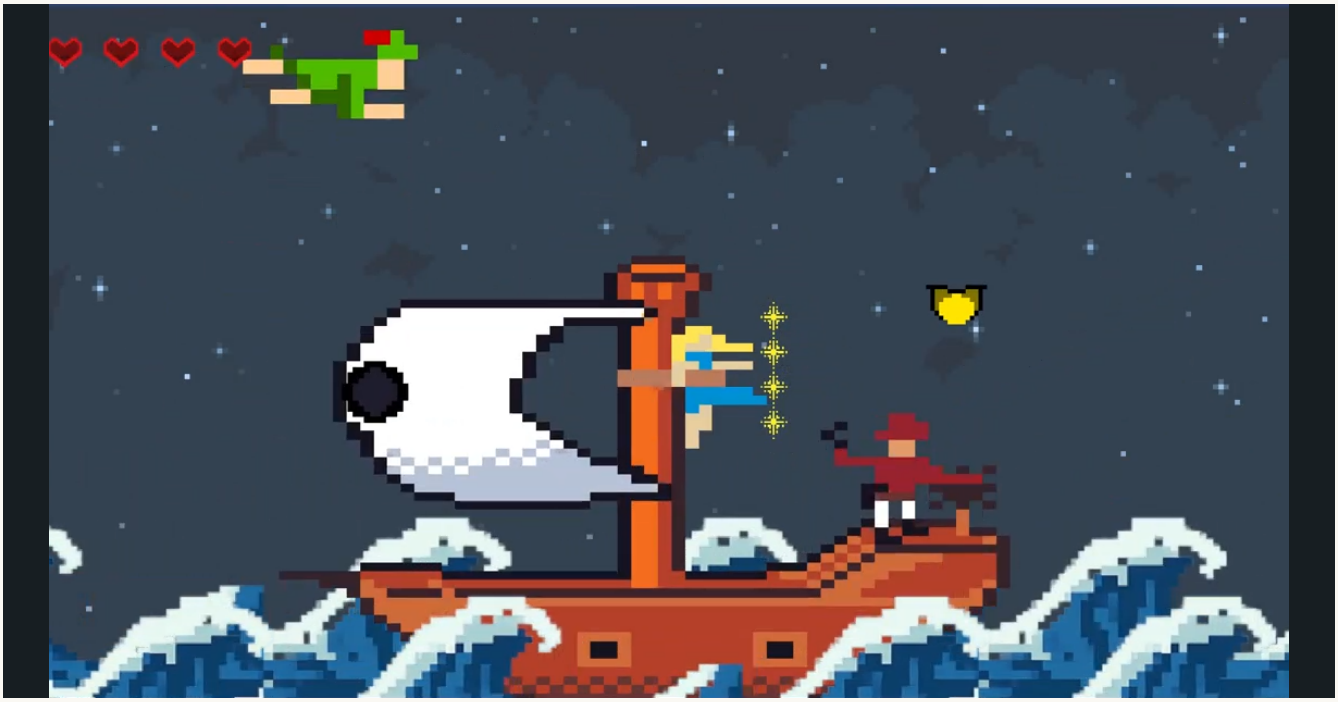
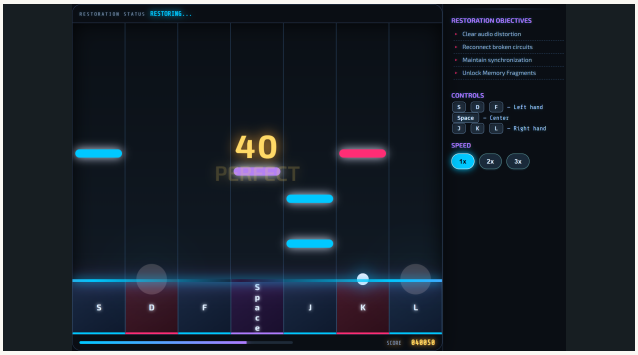
# Changyu Wen

# Game Design Portfolio

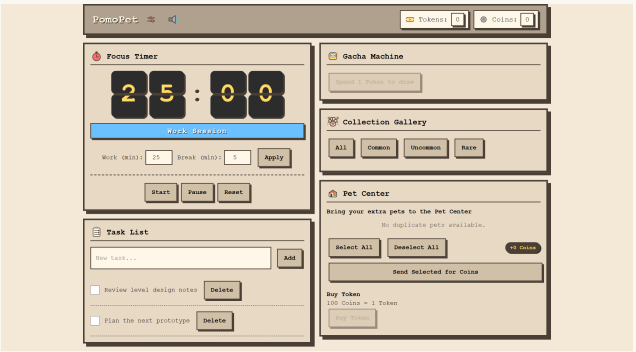
I enjoy seeing people become interested in something I have made. These projects explore how controls, challenge and rewards shape that interest.



PETER PAN / Boss encounter design




FORGOTTEN THINGS / Rhythm interaction




POMOPET / Focus and collection

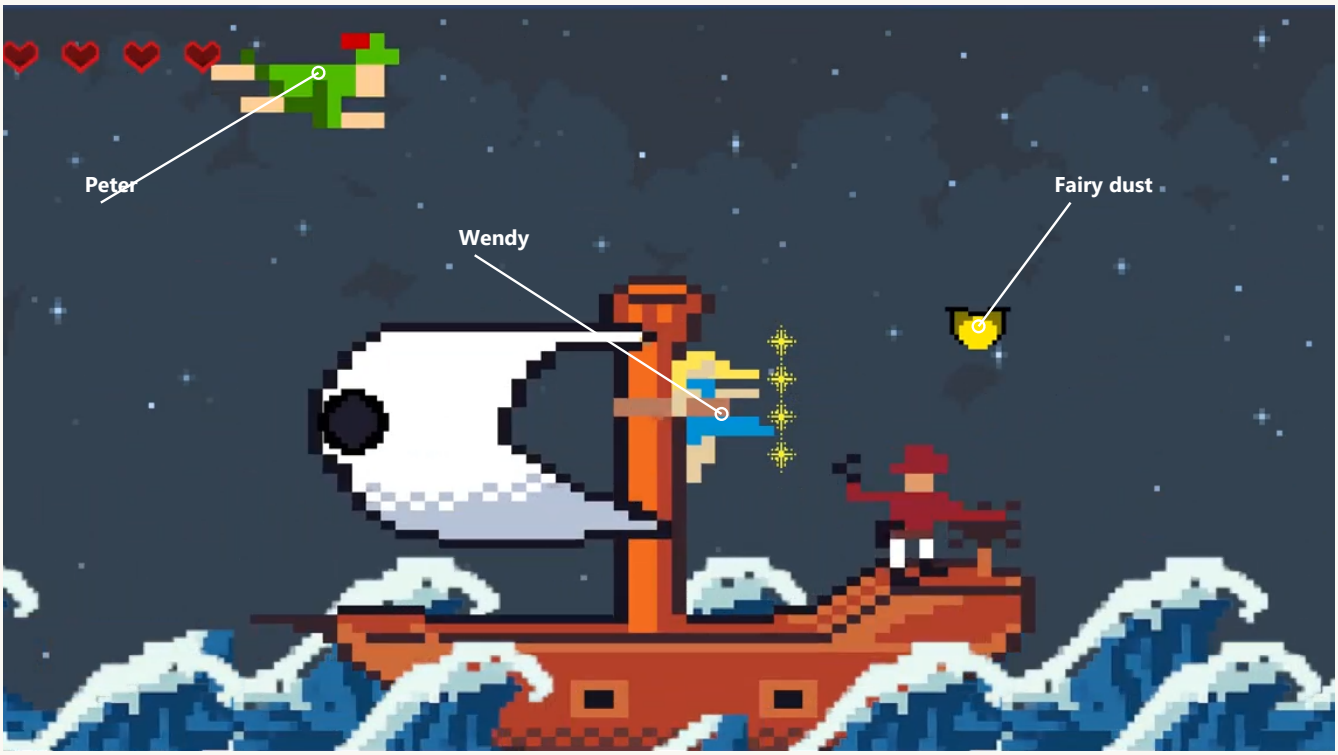
OCT-DEC 2025 / TWO-PERSON PROJECT / UNITY & C#

# Peter Pan

## Escape from Neverland

Watch gameplay (1:37)

Source repository




Touch fairy dust to carry it, then approach Wendy to build rescue progress while avoiding cannon fire.

### THE GAME

A 2D side-scrolling rescue game made with Chris Cox. Peter pursues Captain Hook, collects fairy dust and frees Wendy aboard the ship. Our adaptation gives Peter no attack: the player must survive and rescue rather than defeat Hook.

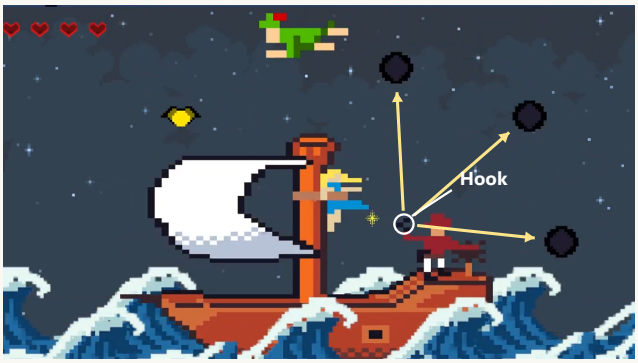
### MY CONTRIBUTION

The non-attacking premise was a joint decision. I designed and developed the Boss encounter, including projectile behaviours, fairy-dust collection and delivery, and rescue progress. I revised the encounter after classroom playtests.

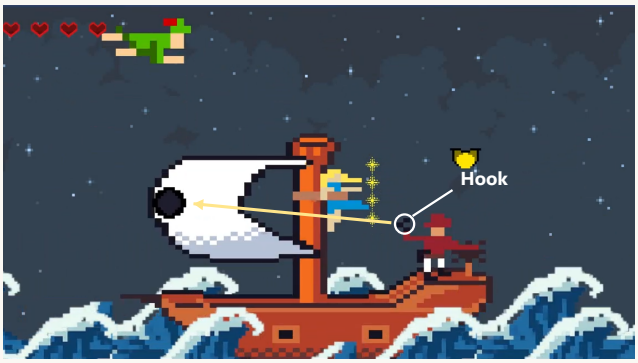
## The design question

How could dodging and rescuing provide enough challenge to sustain a Boss fight?

# Making rescue feel challenging




**Three-shot spread.** Three projectiles fan out from Hook's hook.




**Single aimed shot.** Direction is set at firing; the projectile does not track later movement.

White ring: firing point. Yellow arrows illustrate shot directions.

Aimed shots reward leaving the firing line; spreads ask players to find gaps. Homing shots add pressure to keep moving. This gives the same rescue route different movement demands.

### FIRST PLAYTEST AND REVISION

Single aimed cannonballs were too easy. I kept the rescue objective and random dust placement, then added spread and homing projectiles. The change focused on what players had to dodge while collecting and delivering dust.

### RETEST OBSERVATIONS

Players began looking for gaps between cannonballs and found the difficulty more appropriate. Some still did not know where to take the dust. After I explained the victory condition aloud, they understood the rescue task and adapted.

### THE NEXT TEST

I would test whether players can identify and perform the rescue task without a verbal explanation. The classroom response showed that the mechanic became understandable with guidance.

My role: Boss encounter design and development; overall premise agreed with Chris Cox. AI assisted with projectile code. Third-party UI and audio assets are credited in the game.

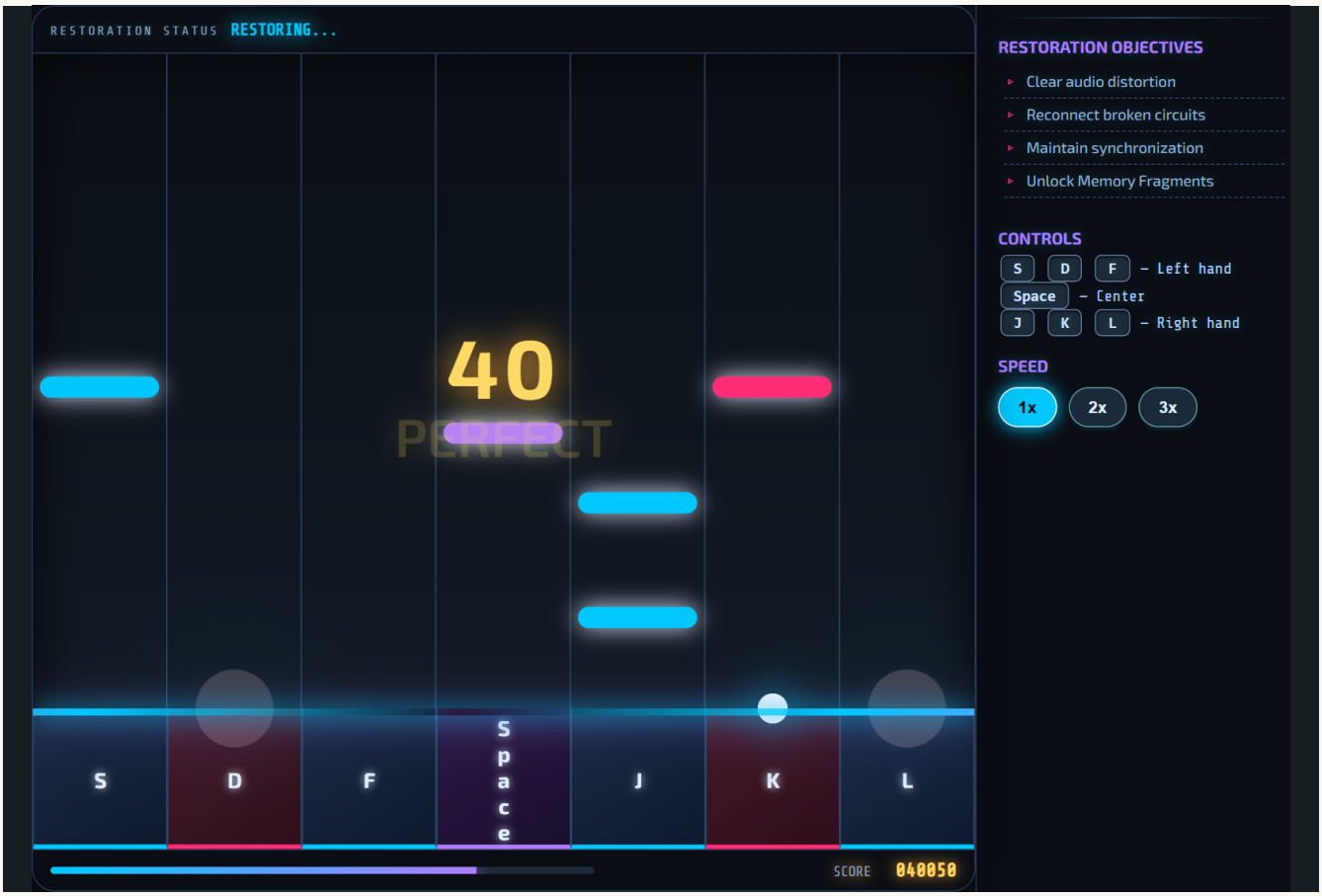
JAN-APR 2026 / SOLO PROJECT / HTML, CSS & JAVASCRIPT

# Forgotten Things

A rhythm prototype about restoring lost music

Play prototype

Watch gameplay (1:01)




Browser prototype: seven lanes, timing judgments, combo feedback and selectable note speed.

## THE EXPERIENCE

The archive setting frames play as recovering forgotten music. The playable version concentrates on the rhythm interface: match falling notes to seven keyboard inputs, with visible timing and combo feedback.

## MY CONTRIBUTION

I developed the concept and design documents, built the browser prototype with AI assistance for timing judgment, and revised the input scheme after a playtest. The broader 3D archive exploration remained a concept.

# Simplifying the first interaction

**EARLY DESIGN DOCUMENT / PLANNED CONTROLS**

| | | | |
|---|---|---|---|
| Rhythm | Left Lane Action | `A` Key | Left side of cabinet pulses with cyan light. |
| Rhythm | Right Lane Action | `D` Key | Right side of cabinet pulses with magenta light. |
| Rhythm | Sky Lane Action | `Spacebar` | Overhead fluorescent lights flicker brighter. |
| Rhythm | Laser Trace | Hold `Left Click` + Move `Mouse` | Glowing wire is drawn in real-time; a "circuit complete" chime plays on success. |
| Rhythm | Fragment Grafting | Flick `Mouse` Left or Right | A specific instrument track becomes audible; screen flashes white. |

Early control specification, Version #1, p. 5: keyboard lanes, mouse tracing and flick gestures.

**WHAT THE PLAYTEST REVEALED**

I had been most excited by the unusual mouse-and-keyboard combination. The playtest made me reconsider it: coordinating different actions distracted from the rhythm. Novel controls were adding work before the player could enjoy the game.

**WHAT I CHANGED**

I removed mouse input and complex note types, and used tap-only input on S, D, F, Space, J, K and L. On trying the revised version, the same player could begin entering the correct inputs without a detailed explanation.

## The next design problem

With the controls easier to understand, I want to develop the impact of each hit through sound and visual feedback, then test whether the interaction remains satisfying over a full song.

My role: concept, design documents, browser prototype and input revision. AI assisted with timing judgment implementation.

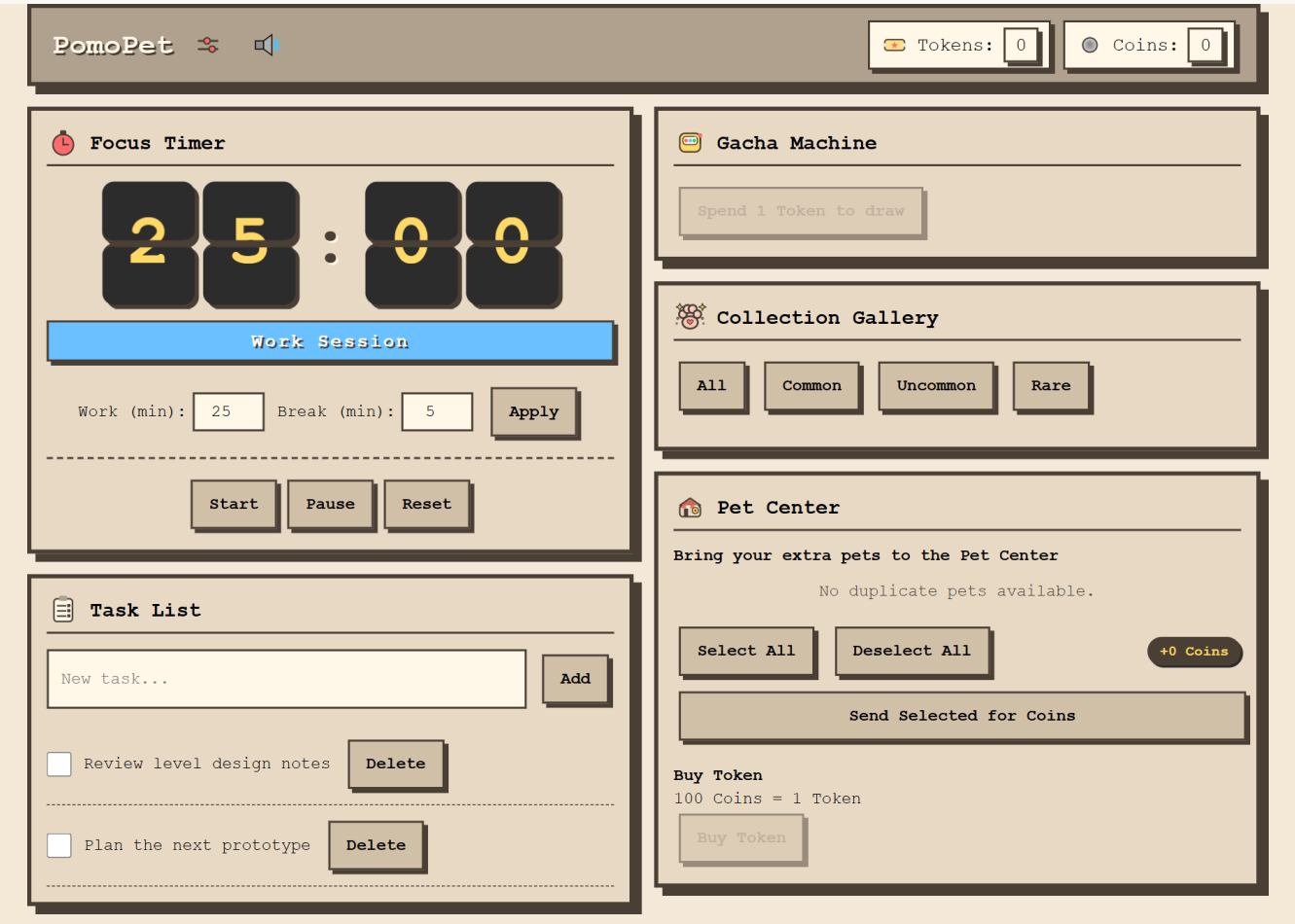
MAR-APR 2026 / SOLO PROJECT / HTML, CSS & JAVASCRIPT

# PomoPet

A focus timer with a pet-collection reward loop

Try PomoPet

Watch demo (0:40)




Timer and tasks sit alongside collection and rewards. Interface shown with example tasks.

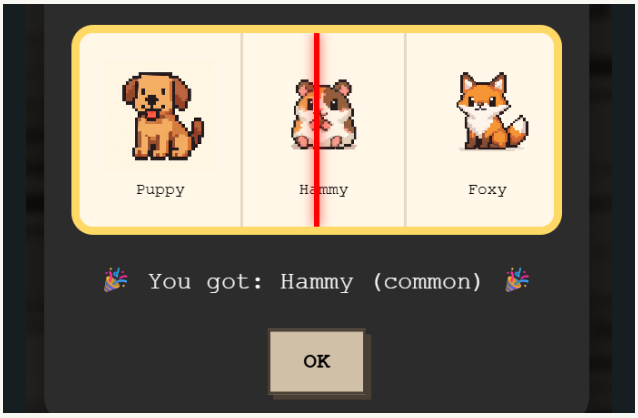
## THE IDEA

I wanted completing a task or focus cycle to give the user something small to look forward to. Each earns a token for a random pet draw. Duplicate pets can be recycled into coins, which can be exchanged for more tokens.

## MY CONTRIBUTION

I wrote the design document before implementation and determined the reward loop, interface requirements and animation behaviour. AI assisted with parts of the JavaScript and generated pet images to my style and content specifications.

# Keeping the reward loop moving

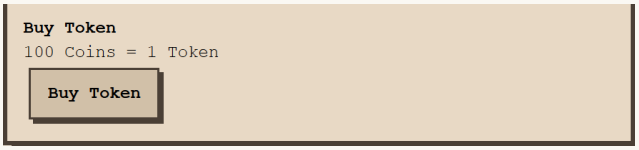



Spend one earned token on a random pet.




Select duplicates, then Send Selected for Coins.

| RARITY | DRAW | RECYCLE |
|---|---|---|
| Common | 75% | 5 coins |
| Uncommon | 20% | 10 coins |
| Rare | 5% | 20 coins |




After recycling, Buy Token exchanges 100 coins for one token.

### 100 coins = 1 token

Current prototype values; further balance testing is needed.

#### ANTICIPATION IN SELF-TESTING

I wanted the rare dragon enough to finish tasks and focus cycles for more draws. Recycling gives duplicate pets a return value, so an unsuccessful draw still contributes toward another chance.

#### WHAT THE LOOP STILL LACKS

A complete collection would leave little game activity. New pets and seasonal collections are possible additions. I also want to test whether collecting remains satisfying when duplicates become more frequent.

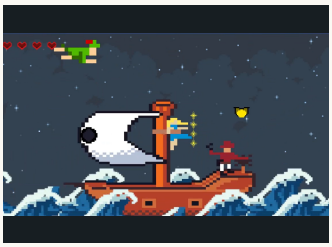
#### DIRECTING THE ANIMATION

I specified horizontal movement, gradual deceleration and a stop on the result. When early AI implementations missed this, I broke the motion into stages and reviewed repeated revisions.

My role: design document, reward rules, interface requirements and animation direction. AI assisted with JavaScript and generated pet artwork to my specifications.

# Explore the projects

Browser prototypes and source repositories



## Peter Pan

Unity / two-person project / Boss design

Watch gameplay (1:37)

Source repository




## Forgotten Things

Desktop browser / keyboard / rhythm prototype

Play the prototype

Watch gameplay (1:01)

Source repository



## PomoPet

Desktop browser / timer, tasks and collection

Try PomoPet

Watch demo (0:40)

Source repository

**CONTACT**

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GitHub profile